Team 5 - Lochinvar Induction Heater Controller

Designed by Aaron Neuharth, John Donnell, Austin DuCrest, Dow Cox, Cole Wilson

Background and Key Stakeholders

Problem Statement: Design and prototype an induction coil controller in such a way to heat both the internal and surface level material of a cylindrical metal pipe.

Key Stakeholders:

- **Lochinvar:** Acts as the customer as well as providing advice.
- **TTU:** Concerned for educational success.

Budget: A combined funding of $2000 ($1k from TTU, $1k from Lochinvar)

Problem Formulation

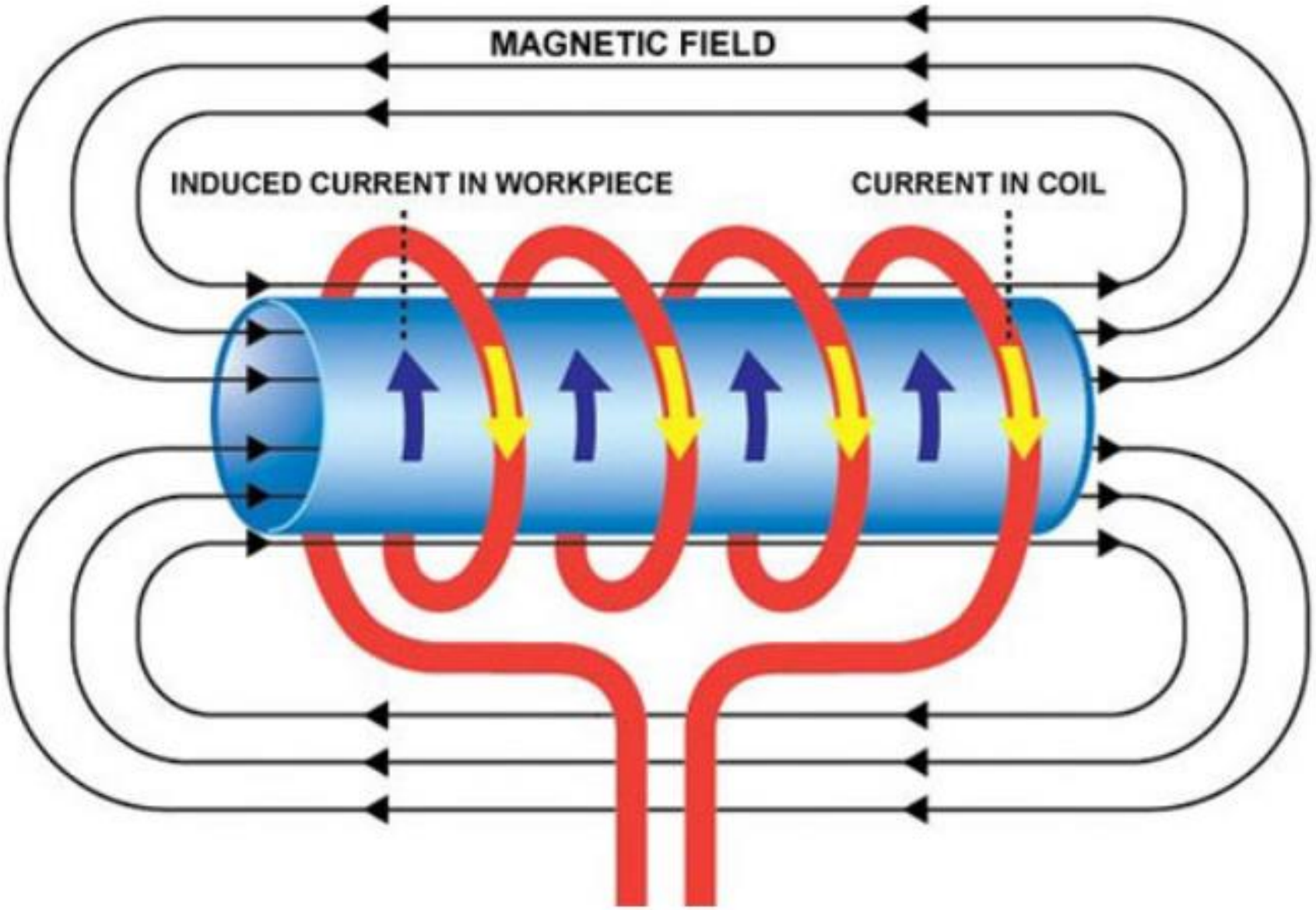


Figure 1. *Induction Heating Principle*. Adapted from "What is Induction Heating and How Does it Work," by HVH Industrial Solutions, n.d., https://hvhindustrial.com/blog/induction-heating-principles

Specifications

1. This controller shall have multiple power settings similar to induction cooktops

2. This controller shall be capable of inducing surface eddy currents and internal heating

3. The controller shall test circular bar stock and standard black pipe thermocouple in such a manner to validate the success of the heating

4. This controller shall measure total electrical power consumed

5. This controller shall measure the total temperature rise of the metal.

Measures of Success

- The controller is capable of inducing eddy currents to validate the success of heating.
- Power level supplied to load is available to user on display.
- Total temperature rise of the metal is measured and outputted to user.

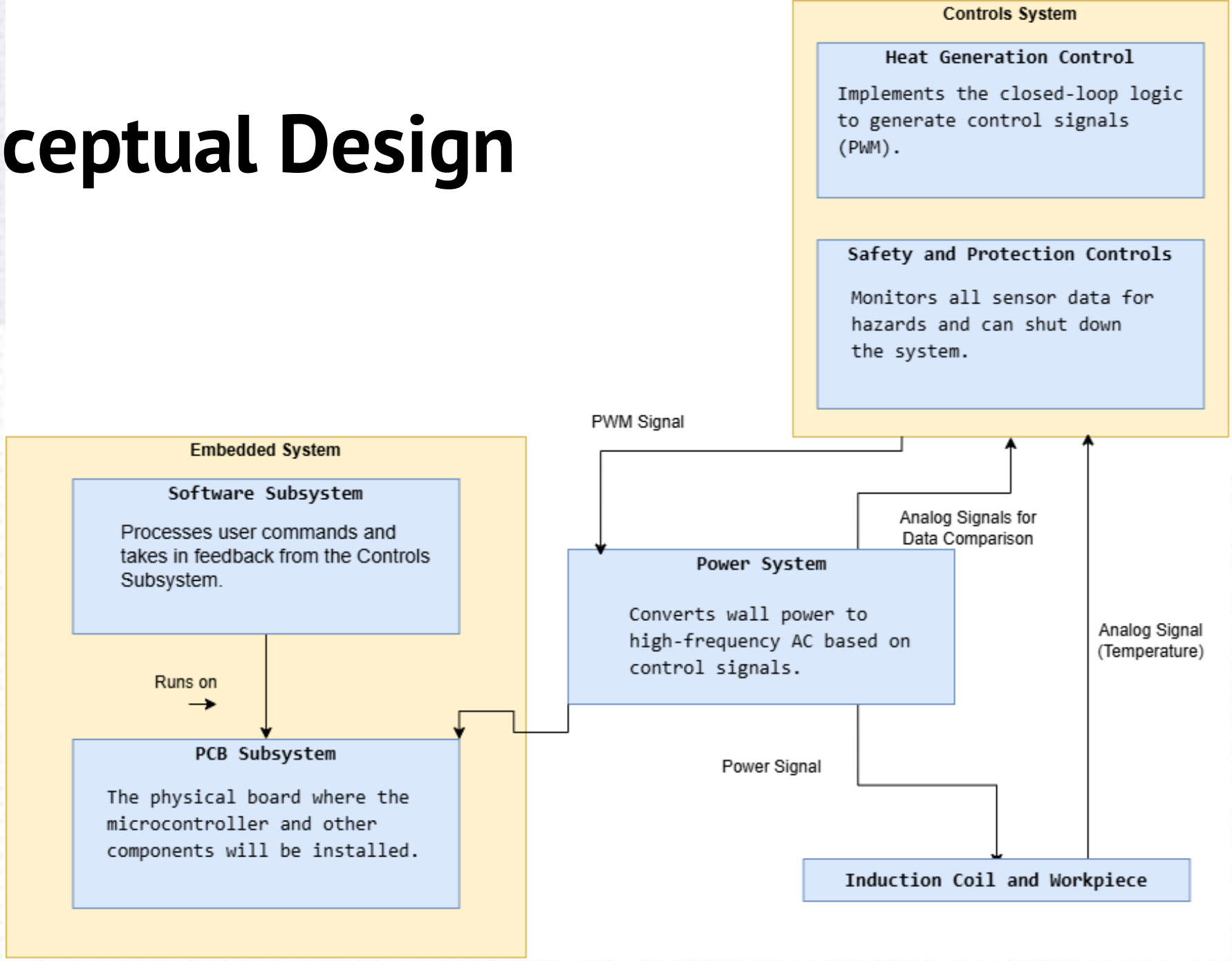
Standards and Regulations

- NEC Article 665: Directs the design of induction heating equipment
- Safety Constraints:
  - An “off” state must open all ungrounded conductors to prevent automatic re-energizing.
  - Control Panel should be dead front construction and any accessible part of the heating equipment should not exceed 50 V

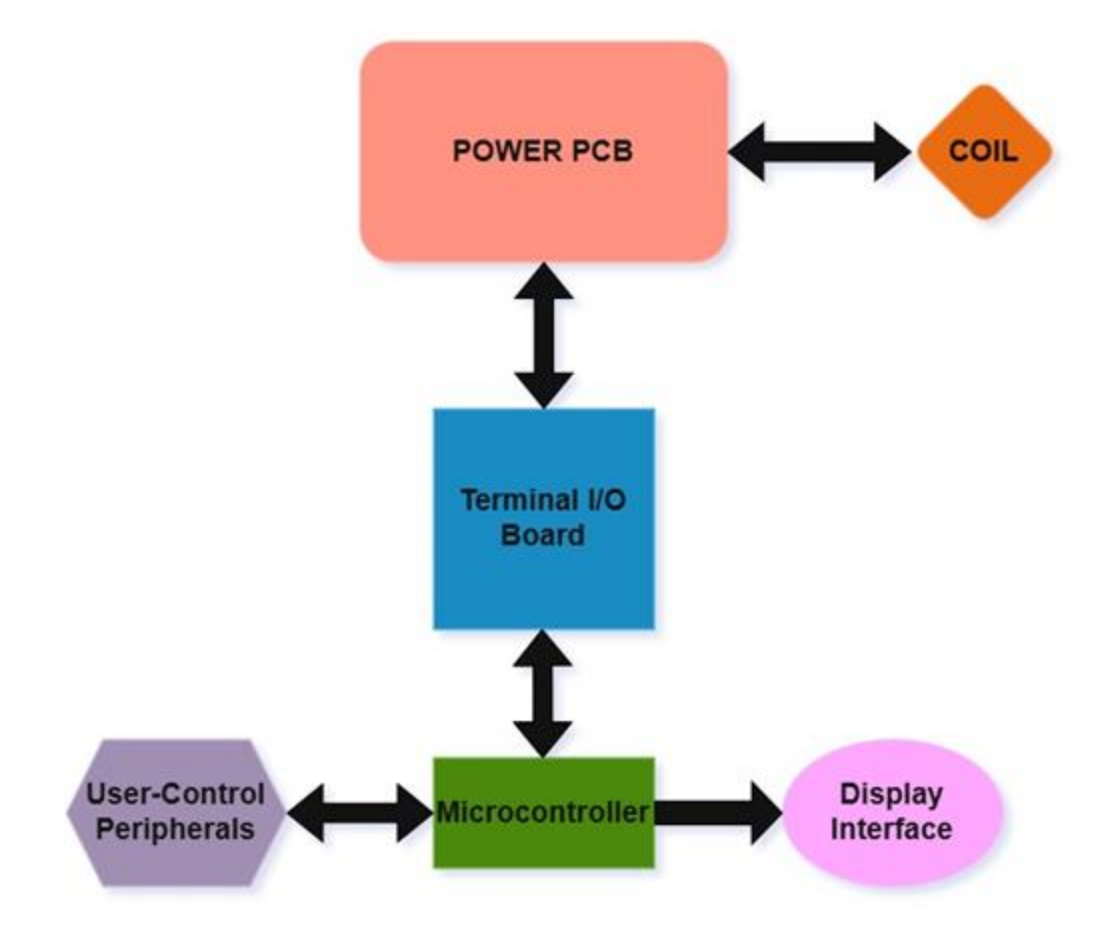
Broader Impacts

- User Safety
  - Use of nonconducting enclosures to mitigate shock and burn risks.
- Thermal Safety
  - Induction heating can reach very high temperatures very quickly, which makes the heat of the pipe a concern.
- Economic/Environmental
  - High-efficiency heating as a cleaner alternative to traditional combustion-based boiler systems.

Conceptual Design



Hardware Block Diagram



- Separate subsystems for power management, I/O interface, user controls, and display
- Power PCB handles power conversion and power transfer
- Terminal I/O Board provides signal conditioning and interface between power and control systems
- Bidirectional communication allows for real-time feedback and system monitoring

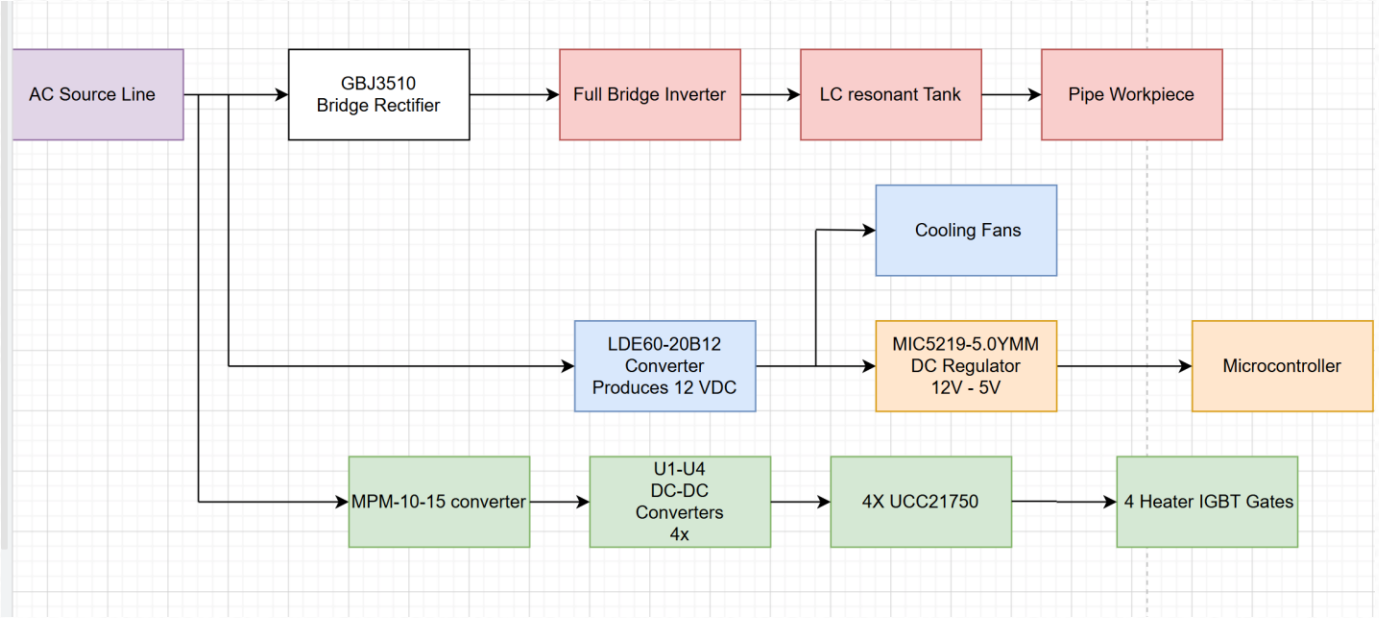
Trade Study

Component	Chosen Option	Alternative(s) Considered	Reasoning
Control Logic	Closed-Loop System	Open-Loop System	More accurate control utilizing thermocouple used to verify temperature rise of the metal
Switching Device	IGBTs	MOSFETs/BJTs	IGBTs are superior for high power applications and have better thermal performance [1][2]
Power Inverter topology	Full-Bridge Rectifier	Half-Bridge Rectifier	More efficient, higher AC voltage output. Handles higher power applications more effectively.
Enclosure Material	Aluminum	Wood/Polycarbonate	More heat dissipation, more durable, and naturally blocks more electromagnetic noise.
Thermocouple	K-type	J, T, E, S	Industry standard, general temp range suitable for this application
Microcontroller	STM32 Nucleo-L476RG	ESP32-C61, DISCO-STM32L4S5I	Familiarity, a lot of each capability ADC, SPI, and PWM for low cost

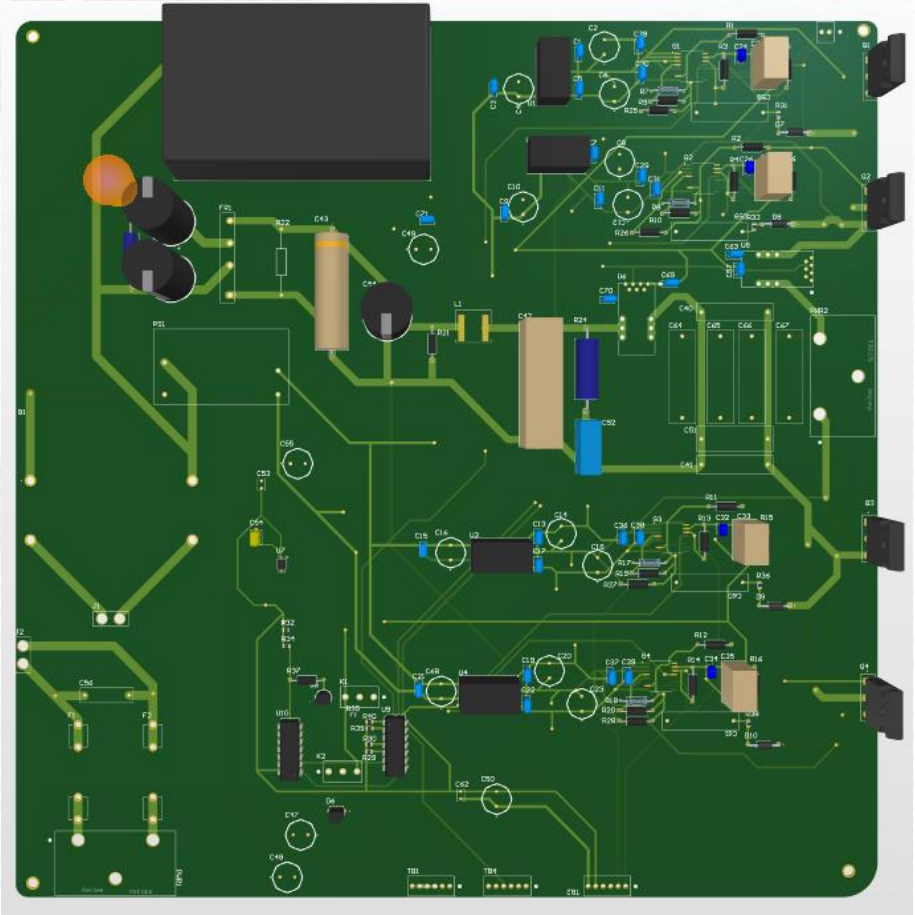
[1] Power Electronics News (2023): "IGBTs for Induction Heaters"
[2] JSW XDH (2021): "IGBT vs. MOSFET: Choosing the Right Semiconductor for High-Power Applications".

Detailed Design - Power

- Proper decoupling and smoothing for input power signals to sensitive components.
- Ensures all components receive proper voltage.
- Components are selected such that they are rated to handle the nominal and spike currents that are generated within system.



Detailed Design - PCB

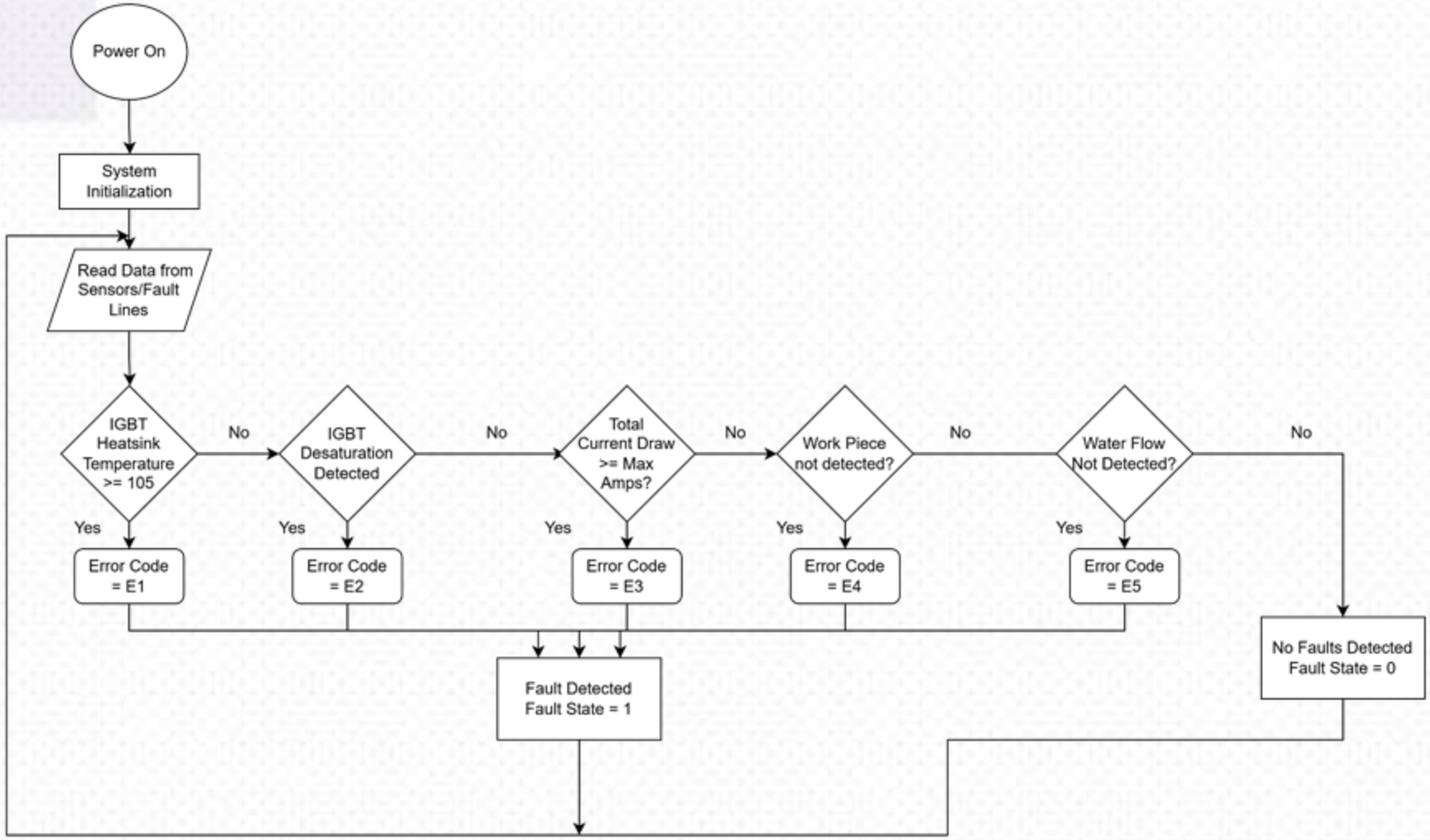


- Supports All Subsystems
- Routes Sensor Feedback Signals
- Interfaces with User I/O Hardware
- Provides Power and Isolation Between Domains
- Carries Fault and Protection Signals
- Organizes and Distributes System Level Connections
- 4 layers, 12X12 board to maintain creepage and clearance of routes

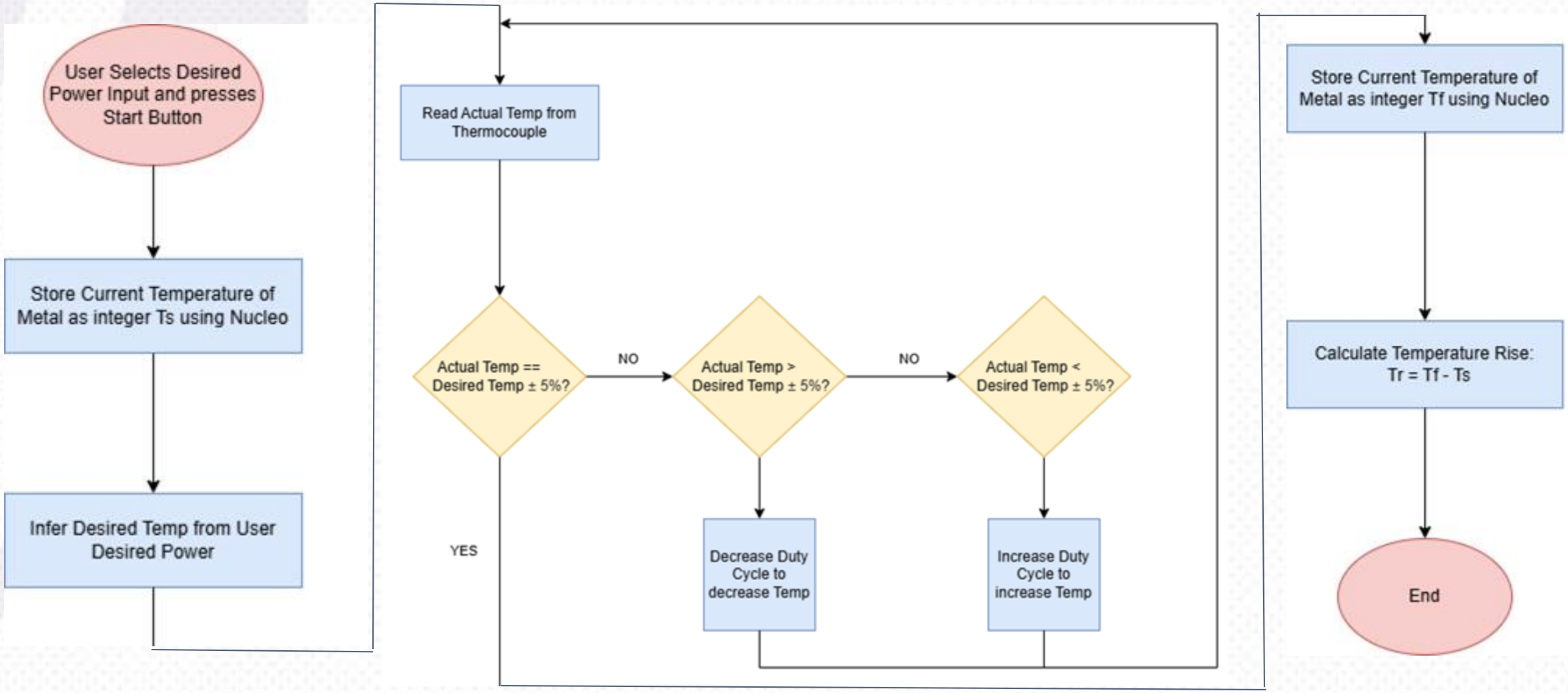
Detailed Design - Safety and Protections

- Hybrid Design between Software and Hardware Safety Interlocks
  - Hardware
    - IGBT Desaturation Circuit
    - Fault Latching Circuit
  - Software
    - Heatsink Temperature Monitoring
    - Input Current Monitoring
- Split Enclosure to isolate Control signals from Power Signals

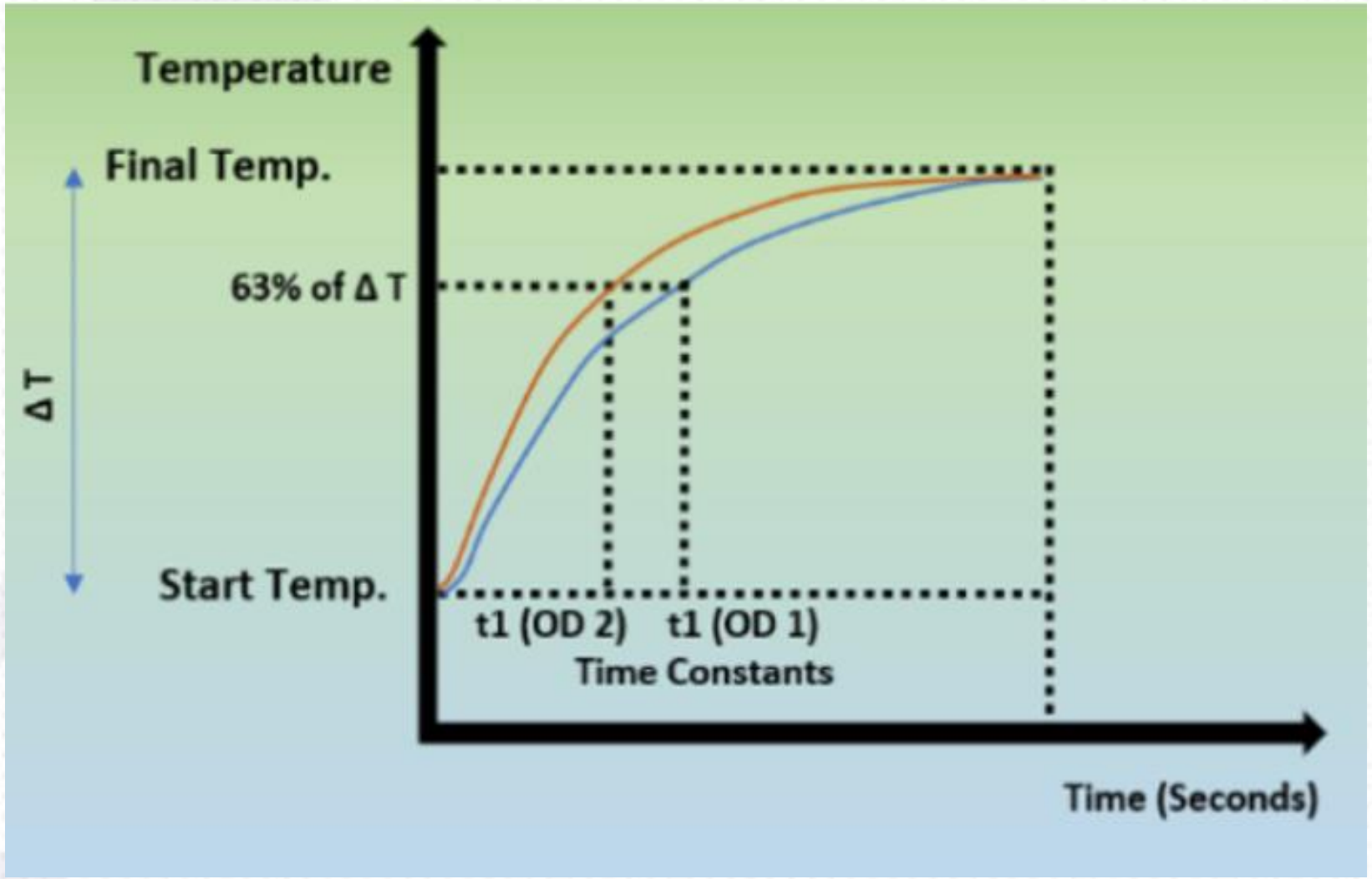
Detailed Design - Safety and Protections

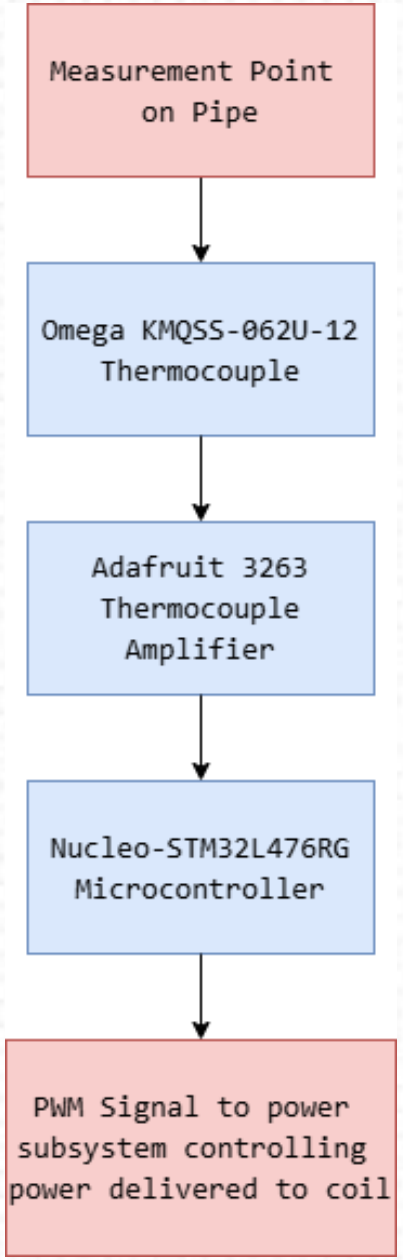


Detailed Design - Heat Generation

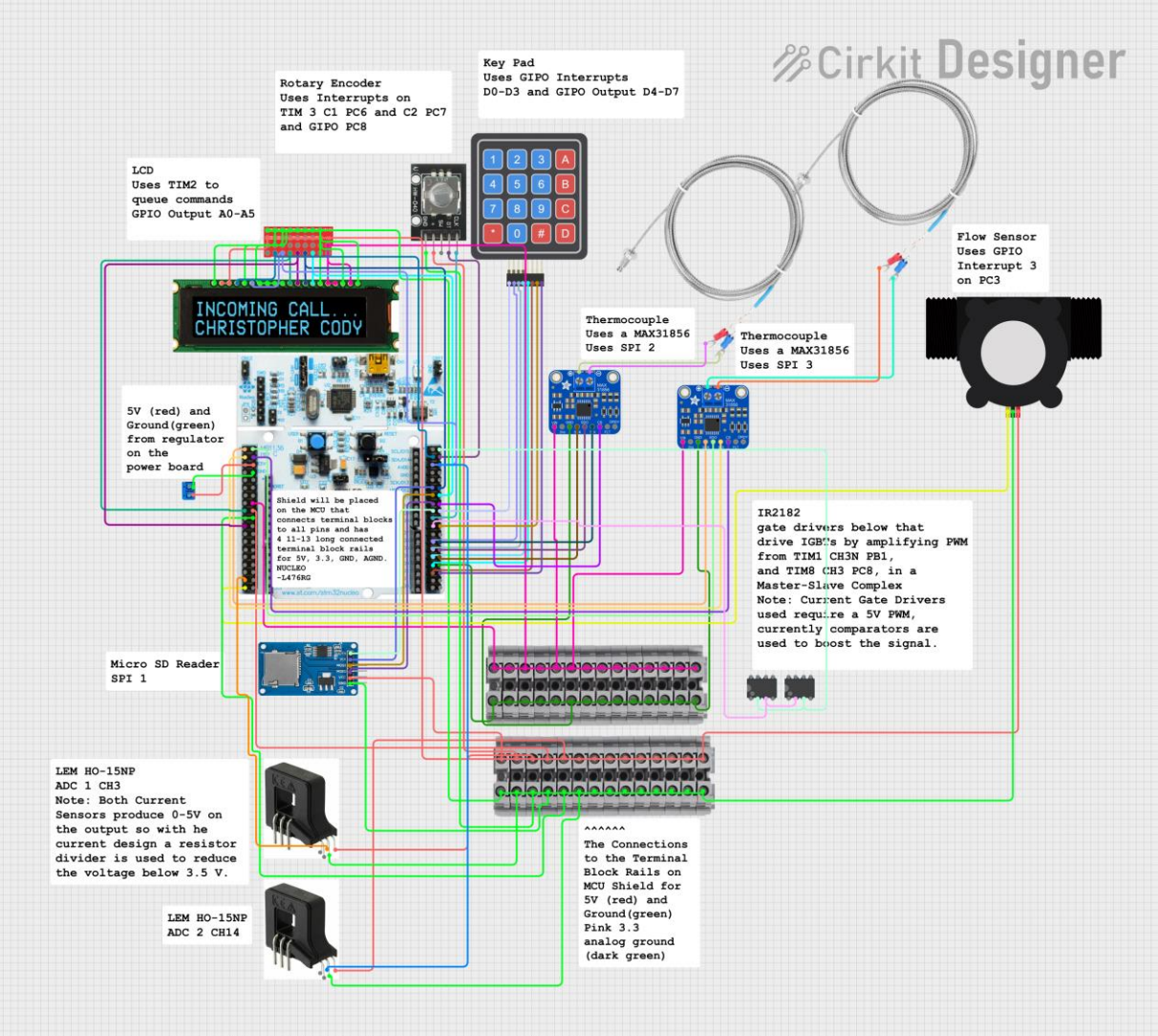


Detailed Design - Heat Generation



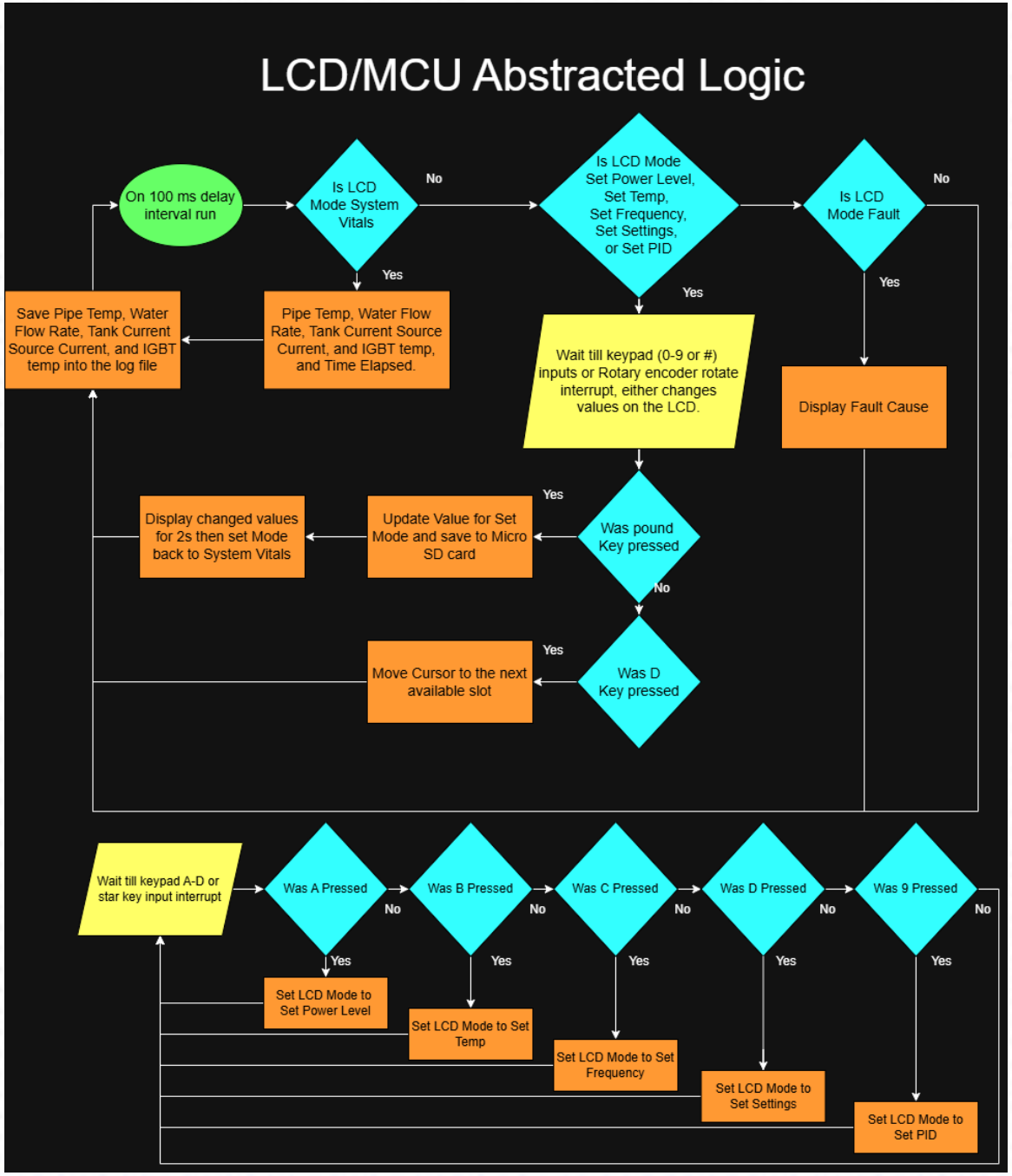


Detailed Design - Embedded



- Connects all Subsystems
- Interfaces with Sensors
- Interfaces with the User
- Runs the Logic for PID Controller
- Displays Faults from Safety System
- Logs System Vials

Detailed Design - Embedded



Budget Overview

- The project succeeded in staying in the expected budget provided by both Lochinvar and Tech.

System Category	Total Combined Cost
PCB Fabrication & Shipping	$500.00
Power Subsystem (IGBTs, Drivers, Diodes, Coils)	$456.53
Capacitors, Resistors, PCB hardware	$222.32
Control & Embedded System (Microcontrollers, Sensors, LCDs)	$183.68
Thermal & Mechanical (Heatsinks, Fans, Enclosures)	$117.78
Total Project Expenditure:	**$1,480.31**

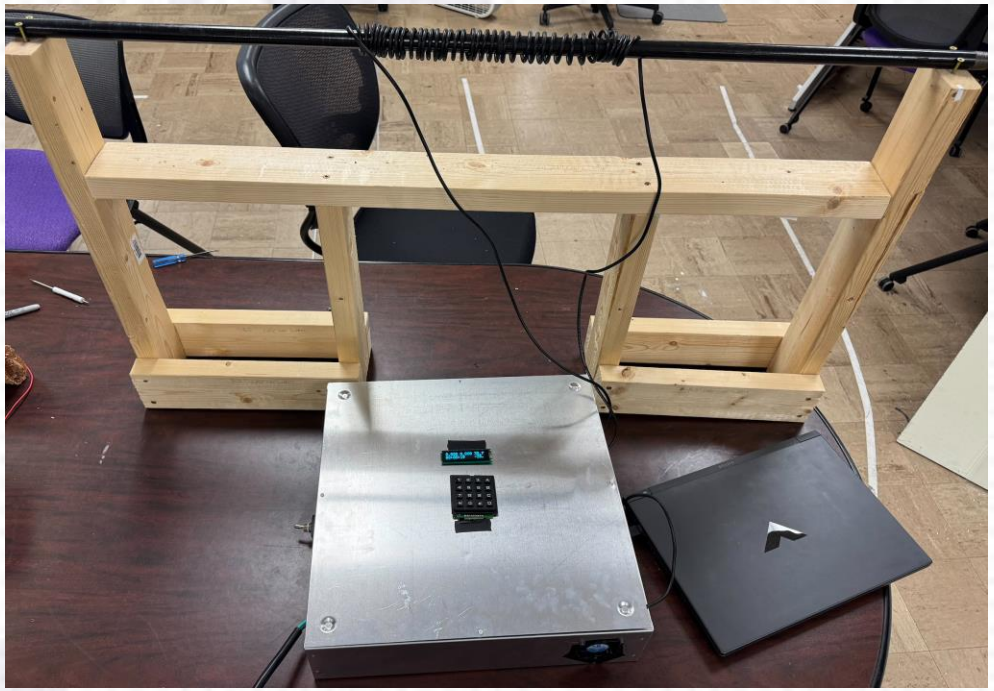
Implementation and Current Status

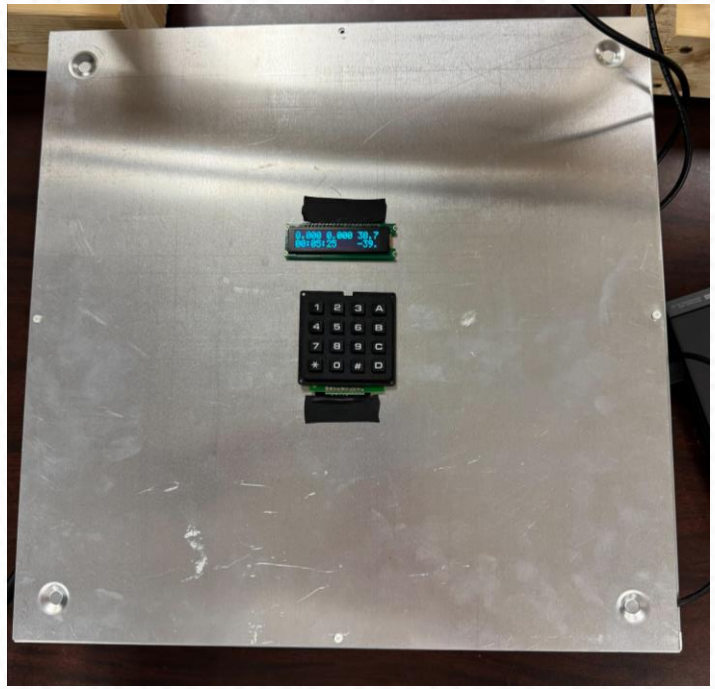
- **Ordering:** Parts procurement was delayed several weeks pending final funding approval.
- **Physical Build:** Due to delayed shipment, a half bridge inverter was utilized instead of a full bridge for test data to analyze.
- **Current State**: Hardware assembly is complete; the system is currently undergoing final continuity and low-power logic verification.

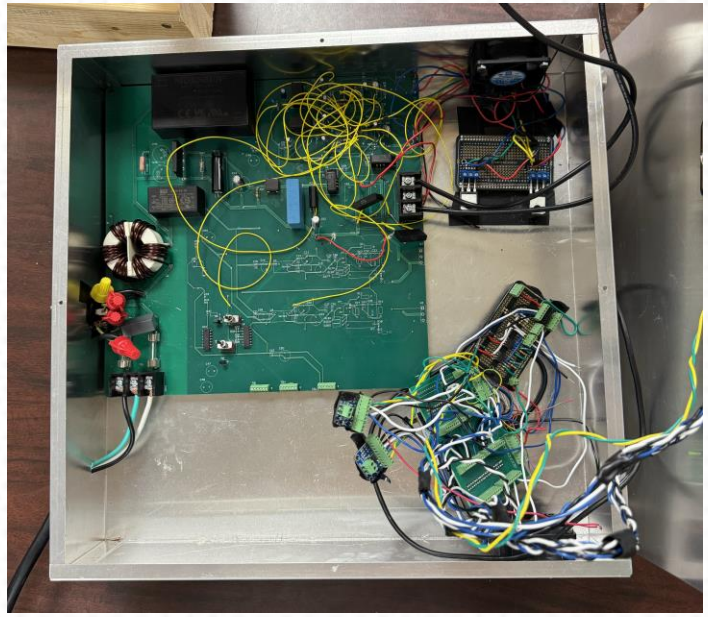
Implementation and Current Status

- Jumper wires and Solder-able breadboards were utilized due to time constraints to overcome issues with the PCB. This workaround was to fix the following:
  - Soldering pads for surface mount parts were placed on an incorrect layer
  - Inconsistencies for the IGBTs were found on the PCB compared to the schematics.

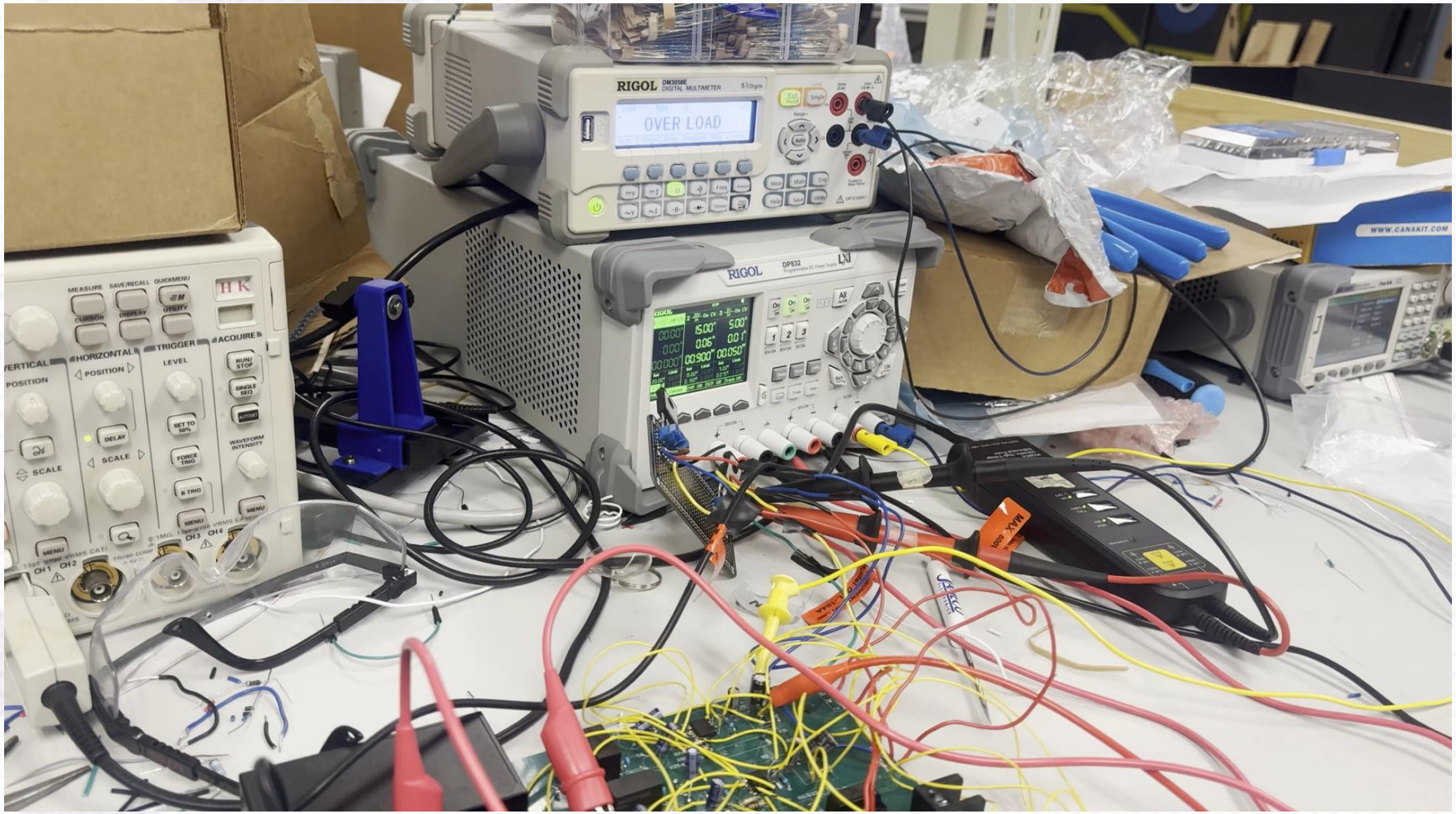
Implementation and Current Status

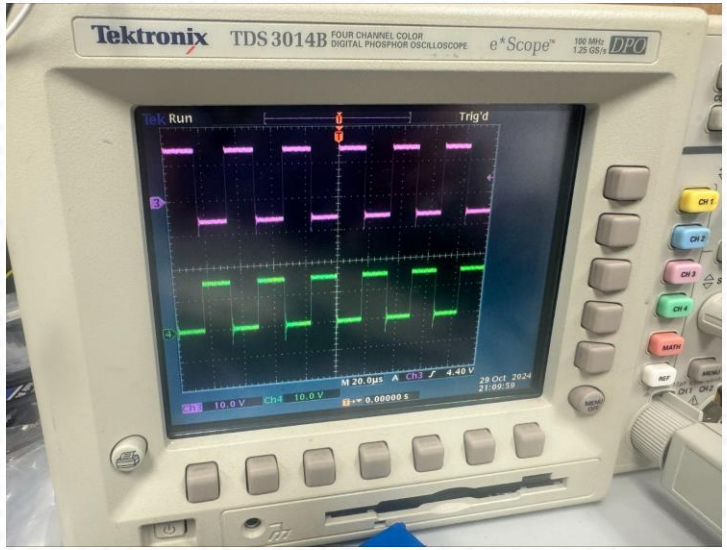


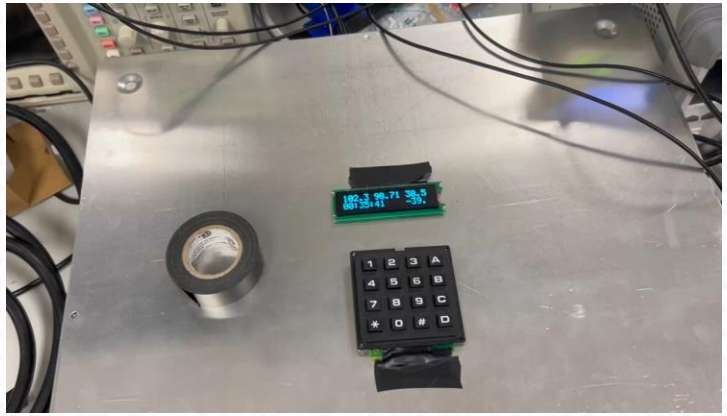




Demonstration







Lessons Learned

- Cross-Team Communication:
  - Ensuring a steady communication between faculty advisor and customer
  - Ensuring the proper financial support is guaranteed and verified a few weeks in advance of ordering is critical to ensure timely ordering and remain within time constraints and deadlines
- Adapting to Unexpected Issues
  - No project works 100% correct first try
  - Ensuring enough time is given to address issues in the prototyping phase of the project is critical

Future Goals

- Full-load testing of the system to measure the thermal rise of the workpiece, going from a 30VDC power supply to a 120VAC supply.
  - Verify and test the effects of a full-load power input from the 120 VAC to improve temperature response times and control
- Fixing the issues with the design of the PCB to correct the traces and the layering of the surface mount components to order a new board and provide a cleaner final product

Statement of Contributions

- Aaron Neuharth - Detailed Design Safety, Conceptual Design
- John Donnell - Detailed Design PCB, Hardware Block Diagram
- Austin DuCrest - Detailed Design Power, Constraints
- Dow Cox - Detailed Design Embedded, Component Temp Constraint
- Cole Wilson - Detailed Design Heat Generation, Lessons Learned

Special Thanks to Mr. Putnam & Mr. Rushing from Lochinvar as well as Dr. Johnson, Mr. Conard Murray and Mr. Chris Mills from TTU.